01 / RESULTS

Clock-Free Cut Quantisation

A two-cut sector theorem, an intrinsic operator cut, and the first glued lattice

The previous cell supplied lawful native transports, retained paths, a dimensionless loop defect and an exact stationary response. This development asks a sharper question: *can a declared observation contract select discrete loop sectors, and can the whole-cell operator then generate its own diagonal cut?* For this finite model, the answer is yes.

Main result. In a rank-two native carrier, compatibility of a determinant-one loop with two fixed noncommuting cuts forces

$$H = \pm \mathbf{1}.$$

For the nontrivial sector $H = -\mathbf{1}$, a four-cycle with unit quadratic link costs has

$$\boxed{\mathcal{L}^2 - 4\mathcal{L} + 2\mathbf{1} = 0, \qquad \lambda_\pm = 2 \pm \sqrt{2}.}$$

The scale-free level ratio is $\rho = 3 + 2\sqrt{2}$. No physical time coordinate or elapsed-time evolution is used.

Inputs and outputs must stay distinct

Inherited machinery	Native cut arithmetic, derived ι, EMK relations, dagger, typed paths, positive coefficient pairing, and the existing proof compiler.
Chosen geometry	Rank two at each vertex; a four-cycle; two independent real central Cayley parameters q, p.
New sector contract	Fix two noncommuting local cut protocols before evaluating the loop; require the loop to preserve each protocol.
Chosen cost	Quadratic transported-record mismatch with unit link weights. These weights determine the absolute operator scale.
Derived results	Central loop sectors; exact spectral projectors; first glued-cell spectrum; root-observer blindness; finite positive response reconstruction.
Open identifications	Physical gravitational variables, continuum constraints, measurement interpretation, and any identification with the actual Riemann ξ source.

What actually changed

The two transport parameters are independent: sector selection does not depend on setting them equal. A one-cut counterexample shows why a single diagonal protocol is insufficient. Two cells are then glued and solved exactly, exposing modes invisible to their shared-root observer.

The RH work contributes a reusable positive-pencil reconstruction and innovation test. Applied here, it reconstructs the finite cell response and terminates after two independent readings. It does not identify this cell with the actual zeta source.

Verification. The unchanged native engine replays 10 serialized symbolic certificates, 354 exact checks and 8 negative controls, including 25 rational parameter pairs. The symbolic identities, rather than the parameter sample, establish the general algebraic formulas.

02 / CONSTRUCTION

Independent transports and loop leakage

Use the inherited native cut-complex field $\mathbb{C}_\Sigma = \mathbb{Q}_\Sigma + \iota\mathbb{Q}_\Sigma$, with $\iota^2 = -1$ and $\iota^\dagger = -\iota$. The finite representation is formed *after* the native construction; a physical Hilbert space is not a primitive input.

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad R^2 = -\mathbf{1}, \quad K^2 = \mathbf{1}, \quad KR = -RK.$$

Here $R^\dagger = -R$ and $K^\dagger = K$. For independent real central parameters q, p, define

$$a = \frac{1-q^2}{1+q^2}, \quad b = \frac{2q}{1+q^2}, \qquad c = \frac{1-p^2}{1+p^2}, \quad d = \frac{2p}{1+p^2}.$$

Thus $a^2 + b^2 = c^2 + d^2 = 1$, and the pairing-preserving transports are

$$U = a\mathbf{1} + bR, \qquad V = c\mathbf{1} + \iota dK.$$

On the oriented four-cycle use $T_1 = U, T_2 = V, T_3 = U^\dagger, T_4 = V^\dagger$. The loop based at the first vertex is

$$H = T_4T_3T_2T_1 = V^\dagger U^\dagger VU, \qquad H^\dagger H = \mathbf{1}, \quad \det H = 1.$$

Q1. Two-parameter holonomy and defect. Native reduction gives the exact identities

$$H = (1 - 2b^2d^2)\mathbf{1} + 2abd^2R - 2\iota b^2cdK - 2\iota abcdRK,$$

$$(H - \mathbf{1})^\dagger(H - \mathbf{1}) = 4\chi\mathbf{1}, \qquad \boxed{\chi = b^2d^2 = \frac{16q^2p^2}{(1+q^2)^2(1+p^2)^2}.}$$

These hold before imposing either cut constraint. The previous one-parameter model is the specialization $p = q$.

Two declared cut protocols

Fix the rank-one self-dagger projectors

$$P_1 = \tfrac{1}{2}(\mathbf{1} + K), \qquad P_2 = \tfrac{1}{2}(\mathbf{1} + \iota R), \qquad Q_j = \mathbf{1} - P_j.$$

They are alternative, noncommuting protocols; this construction does not assign them simultaneous physical outcomes. Under a change of native frame, both the loop and the cuts are conjugated together.

Q2. Exact leakage identities.

$$(Q_1HP_1)^\dagger(Q_1HP_1) = 4a^2\chi P_1, \qquad (Q_2HP_2)^\dagger(Q_2HP_2) = 4c^2\chi P_2.$$

Write $\eta_1 = 4a^2\chi$ and $\eta_2 = 4c^2\chi$. The proof compiler checks these polynomial identities after the nonzero Cayley denominators are cleared. Positivity of the finite pairing makes zero leakage an exact preservation condition.

The variables q, p remain continuously variable before the cut contract. A finite matrix representation alone does not quantise a continuously variable loop parameter.

03 / SECTOR SELECTION

Why two cuts are enough

Q3. Two-cut centrality theorem. Let H be a 2×2 native matrix with $\det H = 1$. If

$$[H, P_1] = [H, P_2] = 0,$$

then $H = \mathbf{1}$ or $H = -\mathbf{1}$.

Proof. The first condition says that H commutes with K, and the second that it commutes with R. The four matrices $\mathbf{1}, R, K, RK$ form a basis of the native matrix algebra. Commutation with both generators eliminates the three nonscalar coefficients, so $H = h\mathbf{1}$. The determinant condition gives $h^2 = 1$. Over the native field of characteristic zero, $(h-1)(h+1) = 0$, hence $h = \pm 1$. □

Specialization to the Cayley family. If $\chi = 0$, then $q = 0$ or $p = 0$, and $H = \mathbf{1}$. If $\chi > 0$, the two zero-leakage conditions force $a = c = 0$. Thus

$$q, p \in \{-1, +1\}, \qquad \chi = 1, \qquad H = -\mathbf{1}.$$

The theorem classifies the compatible sectors. Selecting the nontrivial sector for further study is an explicit choice; no principle here forces nature to choose it.

A single diagonal cut does not suffice

Take $q = 1$ and $p = 1/2$. The exact result is

$$H = -\frac{7}{25}\mathbf{1} - \frac{24}{25}\iota K, \qquad \chi = \frac{16}{25}, \qquad \eta_1 = 0, \quad \eta_2 = \frac{576}{625}.$$

This loop preserves P_1 while remaining noncentral. Varying p with $q = 1$ gives a continuous one-cut-compatible family. Therefore two noncommuting rank-one cuts are minimal for this particular rank-two centrality task. This is not a general minimality claim for all observer algebras.

Compatibility is weaker than literal path equality

At $H = -\mathbf{1}$, the loop preserves every projector by conjugation, while a linear readout $E = (1/2, 1/2)$ satisfies $EH = -E$, not $EH = E$. Replacing cut compatibility by literal equality of path amplitudes would discard the nontrivial sector.

The loop-only catalogue descends through E with induced action -1. The larger catalogue containing both U and V still needs one additional native complex coordinate to predict subsequent readings, as in the baseline memory repair. Memory depends on the admitted operation catalogue.

What can detect the sign?

For a local density matrix, $H\varrho H^\dagger = \varrho$ when $H = -\mathbf{1}$. Nevertheless, the sign is a relative connection around a closed cell: the coherent link-mismatch operator distinguishes it from $H = \mathbf{1}$. This does not make an unreferenced global phase observable.

Imposing these two exact cut constraints on every face would restrict each loop to the centre. The resulting projective holonomy is flat. Richer noncentral curvature requires a broader observation contract, including retained cut defects or additional memory.

04 / INTRINSIC DIAGONAL

The cell generates its spectral cut

On the four vertices, let $x \in \mathbb{C}^8_\Sigma$ and define the unitary covariant cyclic shift by

$$(Sx)_{k+1} = T_{k+1} x_k$$

with cyclic vertex indices. This is a relational reindexing and transport operator; its application does not represent a unit of elapsed physical time.

Declare the dimensionless quadratic cost and its operator

$$\mathcal{E}(x) = \tfrac{1}{2} \|(\mathbf{1} - S)x\|^2, \qquad \mathcal{L} = (\mathbf{1} - S)^\dagger(\mathbf{1} - S) = 2\mathbf{1} - S - S^\dagger.$$

Q4. General cell identity. For the family on page 2,

$$\mathcal{L}^4 - 8\mathcal{L}^3 + 20\mathcal{L}^2 - 16\mathcal{L} + 4\chi\mathbf{1} = 0.$$

Indeed, S^4 contains the based loop at each vertex, so $S^4 + S^{\dagger 4} = 2(1-2\chi)\mathbf{1}$. Expanding $(S+S^\dagger)^4 - 4(S+S^\dagger)^2 + 2\mathbf{1}$ gives that same expression.

Q5. Nontrivial sector: the exact two-level law. For $H = -\mathbf{1}$, all conjugate based loops equal $-\mathbf{1}$, hence $S^4 = -\mathbf{1}$ and $S^\dagger = -S^3$. Set $A = \mathcal{L} - 2\mathbf{1} = S^3 - S$. Then

$$A^\dagger = A, \qquad A^2 = S^6 - 2S^4 + S^2 = 2\mathbf{1}.$$

The native real completion supplies the positive $\sqrt{2}$. Define

$$C_{\text{loop}} = \frac{\mathcal{L} - 2\mathbf{1}}{\sqrt{2}}, \qquad \Pi_\pm = \tfrac{1}{2}(\mathbf{1} \pm C_{\text{loop}}).$$

These are complementary self-dagger projectors, directly verified from $A^2 = 2\mathbf{1}$, and

$$\boxed{\mathcal{L}\Pi_\pm = (2 \pm \sqrt{2})\Pi_\pm.}$$

Since $\operatorname{tr} A = 0$ in dimension eight, each projector has rank four. The spectrum follows by constructing the projectors; no primitive physical spectral postulate is required. The executable certificate proves the rational identity $A^2 = 2\mathbf{1}$, with no floating-point eigenvalue calculation.

Local cuts P_1, P_2	Chosen protocols on the two-coordinate local carrier; they select compatible loop sectors.
Vertex grading Γ	$\operatorname{diag}(\mathbf{1}_2, -\mathbf{1}_2, \mathbf{1}_2, -\mathbf{1}_2)$; bipartiteness gives $\Gamma A \Gamma = -A$.
Spectral cuts $\Pi_\pm$	Derived whole-cell projectors; they resolve the two cost-operator levels.

A ratio survives a common cost rescaling

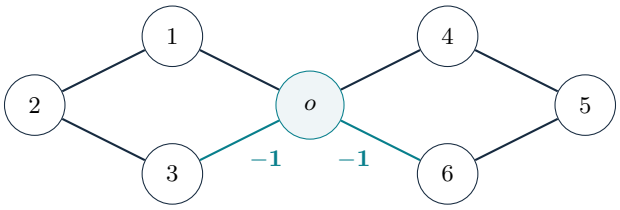
$$\rho = \frac{2 + \sqrt{2}}{2 - \sqrt{2}} = 3 + 2\sqrt{2}, \qquad \rho^2 - 6\rho + 1 = 0.$$

Multiplying every link weight by the same $\kappa > 0$ multiplies both levels by κ and preserves ρ. Unequal weights or graph refinement need not preserve these levels or their ratio. They are dimensionless cost readings, not yet physical energies or masses.

05 / EXTENSION

Two cells: collective and hidden modes

Glue two selected four-cycles at one root. The graph has seven vertices and eight links. In a convenient gauge, each cycle has three $+\mathbf{1}$ links and one $-\mathbf{1}$ link; the local carrier remains rank two.



Unlabelled links carry $+\mathbf{1}$; all link weights are one.

Let D send vertex records to the link differences $x_{\text{target}} - T_e x_{\text{source}}$. The positive cost operator is $\mathcal{L}_2 = D^\dagger D$ on $\mathbb{C}^{14}_\Sigma$.

Q6. First gluing spectrum.

$$(\mathcal{L}_2^2 - 4\mathcal{L}_2 + 2\mathbf{1})(\mathcal{L}_2 - 2\mathbf{1})(\mathcal{L}_2^2 - 6\mathcal{L}_2 + 4\mathbf{1}) = 0.$$

Levels	Multiplicity of each	Sector
$2-\sqrt{2},\ 2+\sqrt{2}$	4	Branch modes invisible at the root
2	2	Antisymmetric branch combination
$3-\sqrt{5},\ 3+\sqrt{5}$	2	Collective root-coupled modes

Proof. It is enough to decompose the seven-coordinate scalar graph and then tensor with $\mathbf{1}_2$. In each branch (a, b, c), the equal-endpoint coordinate $(a + c)/\sqrt{2}$ couples to b, producing a block with diagonal $2, 2$ and off-diagonal $-\sqrt{2}$. These are the two local levels. The remaining coordinate $(a - c)/\sqrt{2}$ couples to the root. The difference between the two branches has level 2; their sum and the root form

$$\begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix},$$

whose polynomial is $\lambda^2 - 6\lambda + 4$. This proves the list and multiplicities. Exact rank checks independently confirm the three grouped kernel dimensions $8, 2, 4$. □

The root cannot see the whole lattice

Eliminating internal vertices gives root static cost $2\mathbf{1}$. For a positive scalar probe t, the root block of the resolvent is

$$r_2(t)\mathbf{1}_2, \qquad r_2(t) = \frac{t+2}{t^2+6t+4}.$$

Only the collective $3 \pm \sqrt{5}$ levels appear. The closure of the root observer under $\mathcal{L}_2$ has rank four, while the full carrier has rank fourteen: ten coordinates belong to sectors invisible to every reading generated by this root protocol.

06 / RH MACHINERY REUSED

A finite response reconstructed exactly

For the single selected cell, Q5 gives the full inverse by multiplication:

$$(\mathcal{L} + t\mathbf{1})^{-1} = \frac{(t+4)\mathbf{1} - \mathcal{L}}{t^2 + 4t + 2}.$$

Its root block is $r(t)\mathbf{1}_2$, with

$$\boxed{r(t) = \frac{t+2}{t^2+4t+2} = \frac{1/2}{t+2-\sqrt{2}} + \frac{1/2}{t+2+\sqrt{2}}.}$$

The positive residues describe this observer. They do not count eigenvalue multiplicities; the full trace of the resolvent is $8r(t)$.

Q7. The SE48 positive-pencil construction reproduces this response. Apply the existing divided-difference construction at probes $t_1 = 1, t_2 = 2$:

$$G_{ij} = -\frac{r(t_i) - r(t_j)}{t_i - t_j}, \quad B_{ij} = \frac{t_i r(t_i) - t_j r(t_j)}{t_i - t_j}, \quad u_i = r(t_i),$$

using derivatives on the diagonal. It gives exactly

$$G = \begin{pmatrix} 11/49 & 1/7 \\ 1/7 & 9/98 \end{pmatrix}, \quad B = \begin{pmatrix} 10/49 & 1/7 \\ 1/7 & 5/49 \end{pmatrix}, \quad u = \begin{pmatrix} 3/7 \\ 2/7 \end{pmatrix}.$$

Both forms are positive: $\det G = 1/4802$ and $\det B = 1/2401$, with positive first diagonal entries. The acquired operator is

$$A_{\mathrm{obs}} = G^{-1}B = \begin{pmatrix} -8 & -7 \\ 14 & 12 \end{pmatrix}.$$

Despite its negative first diagonal entry, it is positive and self-dagger in the acquired pairing G, since

$$GA_{\mathrm{obs}} = B = A_{\mathrm{obs}}^{\dagger} G, \qquad A_{\mathrm{obs}}^2 - 4A_{\mathrm{obs}} + 2\mathbf{1} = 0.$$

Direct inversion gives the rational identity

$$u^{\dagger}(B + tG)^{-1}u = r(t).$$

The executable checks the recorded matrices, pairing relations and exact sample evaluations; the displayed inverse provides the all-t identity away from the poles.

The third reading adds no new direction

At probe 3, write ℓ, j for the new cross-columns of the enlarged G, B matrices. The SE49 innovation quantities are

$$\alpha = G_{33} - \ell^{\dagger} G^{-1} \ell, \qquad \beta = B_{33} - j^{\dagger} B^{-1} j.$$

For this cell, $\alpha = \beta = 0$, and both enlarged forms have rank two. This is exact saturation of a two-pole observer model. It is not the all-order positivity certificate for the actual zeta-source response.

The shared-root response of the glued graph is also a two-pole function, even though that graph has five distinct full-operator levels. Response reconstruction must be paired with an observer-faithfulness or memory-completion argument.

07 / RH CONNECTION

What the diagonal hint establishes

Q8. A finite critical-line pullback. Use the RH adapter coordinate $w = s - 1/2$ and form the *actual determinant of this cell*:

$$X_{\mathrm{cell}}(s) = \det\big(\mathcal{L} + (s-\tfrac{1}{2})^2\mathbf{1}\big) = \big[(s-\tfrac{1}{2})^4 + 4(s-\tfrac{1}{2})^2 + 2\big]^4 .$$

Its zeros are

$$s = \tfrac{1}{2} \pm \iota\sqrt{2-\sqrt{2}}, \qquad s = \tfrac{1}{2} \pm \iota\sqrt{2+\sqrt{2}},$$

each with multiplicity four. Their real part is $1/2$ because the cell operator is positive and the adapter uses w^2. The half-shift is chosen by that adapter; it is not derived from this lattice.

Actual source work already exists

The audited SE48/SE49 branch is further developed than a generic analogy. SE48 records the actual-source identification

$$\mathcal{X}(w^2) = \xi(\tfrac{1}{2} + w), \qquad q_\xi(t) = \mathcal{X}'(t)/\mathcal{X}(t)$$

through its earlier source adapters. Its remaining global gate is all-order positivity, together with the stated source and convergence conditions. SE49 expresses this through successive observer innovations. Those are existing upstream results and conditions; this packet does not re-prove the full branch or rerun its reported numerical certificates.

Our $r(t)$ belongs to a different, explicitly finite source. We have reused the reconstruction laws. We have not shown $r = q_\xi$, matched the cell determinant to ξ, or transferred finite-cell positivity into actual-source positivity at every order.

Symmetry alone is not the missing proof

The real even polynomial

$$f(w) = w^4 + \tfrac{1}{64}$$

obeys reflection symmetries but has the off-axis zero $w = (1+\iota)/4$. The executable verifies this exactly. Thus a reflection or diagonal symmetry does not by itself confine all zeros to the imaginary w axis. The finite-cell result additionally uses a positive operator and a specific determinant.

Repairing the earlier analyticity intuition

The supplied draft *Analyticity as Path Independence Across All Angular Directions* is retained as an uncertified source of intuition. Three distinctions are needed before using it as a theorem:

- Holomorphy alone does not make $\int df/f$ globally path independent. The example $f(z) = z$ on $\mathbb{C}^*$ has period $2\pi\iota$ around the origin.
- An infinite angular sum needs a convergence domain. Taking every angle to be $\pi/4$ already prevents convergence of the partial sums and of their exponentials.
- The equation $\partial_\theta f = \iota f$ selects a specific angular mode; it does not characterize all analytic functions. Symmetry cannot replace the missing positivity argument.

The useful hint survives: track which observations close under an operation, and retain the information that the cut hides. Here it becomes a typed cut-compatibility theorem and an exact positive-pencil calculation.

08 / SCOPE AND NEXT OBJECT

From a cell to an information geometry

A refinement control that fails for a useful reason

Subdivide every original link into two links carrying $(\mathbf{1}, T)$, and assign weight two to each new link. This preserves the total series resistance of each original link and therefore the selected root Schur cost $\mathbf{1}$.

It does not preserve the full cost spectrum. For the resulting sixteen-coordinate operator $\mathcal{L}_{\mathrm{ref}}$, exact native arithmetic gives

$$\operatorname{rank}\left(\mathcal{L}_{\mathrm{ref}}^2 - 4\mathcal{L}_{\mathrm{ref}} + 2\mathbf{1}\right) = 16.$$

In particular, neither old level is an eigenvalue of this refined operator. A static boundary invariant can survive while internal operator levels change. The cell gap is therefore not yet a refinement-independent or continuum physical gap.

Present claim boundary

Established here	Conditional sector selection, exact finite operator spectra, scale ratio under uniform rescaling, gluing, memory-dependent visibility, and finite response reconstruction.
Chosen	The local representation, graph, two-cut preservation contract, selected $-\mathbf{1}$ sector, positive pairing, quadratic cost and weights.
Not yet derived	A physical metric/coframe, gravitational source law, physical quantum measurement law, continuum constraint closure, or identification of cost levels with masses or energies.
RH status	A finite determinant has critical-line zeros. The actual ξ source remains subject to its own global positivity and faithfulness gates.

The next mathematical object

The gluing calculation identifies what to retain next: a full boundary response operator together with its generated observation closure. For a boundary B and internal vertices I, the clock-free elimination law is

$$K_B = \mathcal{L}_{BB} - \mathcal{L}_{BI}\mathcal{L}_{II}^{-1}\mathcal{L}_{IB},$$

when the internal block is invertible. Its probe-dependent extension uses the same Schur complement after adding $t\mathbf{1}$. The next construction should compare these boundary operators under gluing and refinement while explicitly carrying modes that a restricted observer misses.

For noncentral curvature, retain η_1, η_2 and the associated memory rather than imposing that both vanish on every face. This keeps the exact selected sector as a tractable special case, while avoiding a premature restriction of the whole geometry to central holonomy.

Why time can remain deferred

Path composition, loop comparison, cost minimization, positive response and observer closure are all defined without elapsed time. They provide a static relational foundation. A later clock would require an explicitly identified clock subsystem or evolution relation; it should be added as a new construction with its own conditions.

Research consequence. One dimensionless scalar can certify a loop property, but cannot generally reconstruct a lattice. The full response, the admitted operations and the hidden-memory certificate must travel together. This is the concrete extension supported by the first gluing result.

09 / PROVENANCE AND REPLAY

What was reused and what was checked

This is a companion to *Clock-Free Native Lattice*, whose complete source packet is included unchanged under `baseline/`. Source reuse avoids presenting existing framework machinery as newly discovered mathematics. The audit is targeted, not an exhaustive novelty search across every manuscript.

Native foundation	RKF main, commit `e8e3089745bf`: unchanged native arithmetic and Paninian compiler, plus baseline sources for clock-free composition, EMK curvature, memory repair and source response.
T21 / T22	Cut–memory spectral representation and the source/recognized/memory faithfulness gates. These do not license an unseen complement or automatic positivity.
SE26	Diagonal cut and circle transport context. Retained as a source; not a claim that every upstream subsection was independently verified here.
SE48 / SE49	Positive spectral pencils, observer innovations and actual-source RH boundaries; branch head `4faee4f9467e`.
Native gauge	Native positive pairing, compact-group adapters and loop readout; Publications head `7b34bc657883`. No unique physical group or coupling is inferred.
Analyticity draft	User source, retained as an uncertified draft. Its failed implications and useful intuition are separated on page 7.
This assembly	Independent-parameter two-cut selection; intrinsic four-cycle spectral cut; explicit two-cell gluing and blindness calculation; exact finite-response specialization.

Verification layers

- **Native symbolic replay:** 10 serialized certificates in three declared rewriting presentations. Rules include source or model-contract annotations; confluence and dagger gates are audited.
- **Exact arithmetic checks:** 354 checks including 25 rational parameter pairs, operator polynomials, ranks, Schur complements, cut leakage, observer forms and innovations. No numerical eigensolver or independent scalar engine is used.
- **Negative controls:** 8 explicit boundaries, including a noncentral one-cut-compatible loop, an off-axis symmetric-polynomial zero, finite-source saturation, refinement sensitivity and hidden glued modes.
- **Human-readable proofs:** General centrality, projector construction and gluing decomposition appear in this note. Native certificate replay is not a claim of Lean formalization, external peer review, or physical validation.

Reproduce

Unzip the companion packet and run `python3.12 verify.py`. Python checks source integrity and invokes `node verify_quantisation.cjs -check`. Mathematical arithmetic remains inside the unchanged native implementation. The check is read-only and compares the entire freshly replayed JSON result to the recorded result.

`SOURCE_PINS.json` contains full commit URLs and Git blob identifiers. `MANIFEST.json` supplies SHA-256 hashes for every packaged file except itself. `QUANTISATION_RESULT.json` contains the certificates, exact matrices, checks and counterexamples; `SOURCE_AUDIT.txt` records the boundaries of source reuse.

Result status: exact finite, dimensionless, clock-free operator development under declared assumptions. Its physical-gravity and actual-zeta identifications remain research tasks.